

Make biological decision models usable and testable

| For agent developers and researchers

| Aim: economical repeated decisions and adaptation

| Built: traceable inputs, models and learning results

Lower power and an advantage over ordinary AI remain hypotheses.

Record the environment and stimuli in TXs

| Initial TX: dimensions, seed and hazards

| Stimulus TX: one confirmed positive input adds one food

| Open the environment and food source TXs on screen

All external inputs to the flies come from onchain data.

Turn the confirmed world into movement or rest

- | Verify TXs and events before applying inputs
- | Encode the environment and internal body state
- | 7 measured neurons, 19 connections → action → outcome

Encoding, body state, decisions and rendering run offchain.

Replay experience and compare the candidate

- | Select an agent and “Start learning”
- | Update Q-values from a confirmed-world copy and experience
- | Higher comparison score: adopt / Otherwise: keep

Same-environment replay is separate from unseen-world testing.

Separate working integration from improved behavior

Public: a scheduled TX increased food from 3 to 4

Video: four full agents, 2 Aqua and 25 V3 settlements

Research: reward 18.79 → 56.12; direct-input control 56.08

Integration and learning work; biological superiority is unproven.

Measure useful decisions within a small budget

- | Compare small AI models on the same inputs and task
- | Measure reward, failures, adaptation, RAM and joules/action
- | Apply to Uniswap execute/hold and Aqua offer/withdraw

Lower operating cost could encourage adoption; that is unmeasured.

Where code runs and what crosses each boundary

Boundary	Language / runtime	Responsibility	Data crossing it
Browser	JavaScript / HTML / CSS	Public: 7 neurons + Q-learning Full mode: controls + display	HTTP / JSON Verified inputs / view state
Read gateway	JavaScript Cloudflare Workers	Verify receipts and events Anvil shares the read logic	JSON-RPC Blocks, logs, receipts
Signer	JavaScript + ethers Worker / wallet / Anvil	Owner-authorized TXs Budget-capped scheduled sends	ABI / signed TXs No server key returned to UI
Local controller	JavaScript / Node.js	Full world, rewards, execution Ordinary code compares routes	HTTP / JSON ↔ UI JSON Lines ↔ Python
Full-neuron runtime	Python / NumPy / SciPy	166,700 neurons + readouts Foraging selection / market updates	Numeric inputs → actions Outcomes → learning / storage
EVM contracts	Solidity 0.8.30 Project-owned contracts	Identity, model refs, world inputs Owner checks / market settlement	ABI / events Chain state / successful receipts

Shared types: a TypeScript proposal. Policies stay in browser/local storage. Full-neuron computation does not run in Workers.